Country Comparison: Aura Health Hub

Which national systems achieve the five capabilities, and which do not | **1** Unified EHR **2** Multi-tenancy **3** ETL / data standardisation **4** Live audio translation **5** Speaker diarization / ambient scribe

#	Country / System	Year	Approach / Technique Used	Points Achieved	Points Not Achieved
1	**Estonia - EHIS, X-Road**	2024	HL7 CDA R2 documents mandated for all providers since 2009; X-Road exchange; migrating to FHIR with FHIR Mapping Language	**1 Unified EHR:** mandatory national record for all citizens. **3 ETL (partial):** CDA-to-FHIR transformation of structured documents.	**2:** tenant design not published. **3:** no ETL for Word, Excel or PDF. **4, 5:** nothing found.
2	**Finland - Kanta Services**	2016	Central HL7 CDA R2 archive accessed via local systems; legacy-data archive; My Kanta personal health record on FHIR	**1 Unified EHR:** national archive and prescriptions. **3 ETL (partial):** legacy archive and CDA/FHIR standards.	**2:** not published. **3:** no unstructured-file pipeline. **4, 5:** nothing found.
3	**England - NHS**	2025-2026	Summary Care Records for emergency access; NHS England ambient scribing guidance, supplier registry (19 suppliers), template DPIA, MHRA guidance (Jul 2026)	**1 Unified EHR (partial):** emergency summary records. **5 Scribe (partial):** national governance and registry for ambient scribes.	**1:** no full longitudinal national record. **2, 3, 4:** nothing found. **5:** scribes are separate products; diarization not singled out.
4	**United States - TEFCA, Epic**	2026	Federated network-to-network exchange via Qualified Health Information Networks with FHIR-based exchange; Epic ambient-voice specifications	**1 Unified EHR (partial):** federated national exchange. **3 ETL (partial):** FHIR-based exchange. **5 Scribe (partial):** vendor-level ambient AI.	**1:** no single record. **2:** not published. **4:** nothing found. **5:** not a national programme.
5	**UAE - Riayati, Malaffi, Nabidh**	2023-2025	Federal record linked to Abu Dhabi and Dubai exchanges; FHIR-based integration; Malaffi: 3.5 billion records, 12.7 million patients	**1 Unified EHR:** nationally linked record. **3 ETL (partial):** FHIR-based integration.	**2:** not published (regional separation is the closest analogue). **4, 5:** nothing found.
6	**Saudi Arabia - Nphies, Sehhaty, Seha Virtual Hospital**	2025	National exchange platform (stated goal of 100% coverage); citizen access app; telehealth with AI-assisted imaging	**1 Unified EHR (partial):** national exchange and citizen access in rollout. **3 ETL (partial):** standardised exchange.	**1:** unified record still described as a goal. **2, 4, 5:** nothing found (AI used for imaging only).
7	**Jordan - Hakeem**	2025	Single operator (Electronic Health Solutions) runs one shared EHR; 482 public facilities connected	**1 Unified EHR (partial):** shared record across public facilities.	**1:** private sector excluded. **2, 3, 4, 5:** nothing found.
8	**Egypt - Universal Health Insurance**	n.d.	Standardised EMR systems piloted in UHI governorates, linked through a digital backend	**1 Unified EHR (partial):** pilots in rollout governorates.	**1:** no nationwide record yet. **2, 3, 4, 5:** nothing found.

'Nothing found' means no public evidence was found in the sources reviewed, which does not prove absence. England's entry for point 1 follows the project's own research (Summary Care Records, no full longitudinal record).

Summary: Who Achieved What

Country	1. Unified EHR	2. Multi-tenancy	3. ETL / standardisation	4. Live audio translation	5. Speaker diarization / ambient scribe
Estonia	Achieved	Not published	Partial	Not found	Not found
Finland	Achieved	Not published	Partial	Not found	Not found
England (NHS)	Partial	Not published	Not found	Not found	Partial
United States	Partial	Not published	Partial	Not found	Partial
UAE	Achieved	Not published	Partial	Not found	Not found
Saudi Arabia	Partial	Not published	Partial	Not found	Not found
Jordan	Partial	Not published	Not found	Not found	Not found
Egypt	Partial	Not published	Not found	Not found	Not found

Achieved = delivered at national scale. Partial = in rollout, limited in scope, federated, or vendor-led. Not found = no public evidence. Not published = tenant-isolation design is rarely disclosed by national programmes.

Selected Approach for Aura Health Hub

No reviewed system achieves all five points. Points 2 and 4 are not achieved by any country, and point 5 is reached only partially, in England and the United States. Based on what the successful systems share, a hybrid, standards-based, multi-tenant approach was selected for Aura Health Hub:

- **Point 1:** Estonia, Finland and the UAE show that a unified record works through mandated standards and a governed shared layer, without replacing local systems. Aura will use FHIR as its canonical model and a shared patient index.
- **Point 2:** no country publishes a tenant model, so Aura will define one: isolated tenants per hospital or clinic, linked in the way the UAE links regional exchanges to a national layer.
- **Point 3:** every system standardises at source and none ingests Word, Excel or PDF files. Aura will add an ETL pipeline for these sources with human validation, keeping original files as Finland keeps its legacy archive.
- **Point 4:** no national precedent exists. Aura will treat Arabic-first live translation as an additional feature, considered during the design phase.
- **Point 5:** England's governance (patient notice, clinician review) is the model. Aura will build speaker diarization and note drafting into the record, with every AI output reviewed and approved by a clinician before entering it.